Hierarchical Coordination in Cooperative Multi-Agent Reinforcement Learning: A Comparative Review of ALMA and HC-MARL with a Search-and-Rescue Framework and Implementation

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***Abstract*—Hierarchical multi-agent reinforcement learning (MARL) has emerged as a natural response to coordination problems involving partial observability, long horizons, and large agent populations. However, hierarchy can be introduced in different mathematical forms, and different designs solve different coordination bottlenecks. This report presents a comparative review of two representative methods: ALMA, which introduces hierarchy through explicit subtask allocation followed by low-level decentralized execution, and HC-MARL, which introduces hierarchy through multi-timescale consensus learning under centralized training with decentralized execution. We first review the theoretical basis of these two methods using a common Dec-POMDP and hierarchical reinforcement learning framework. We then analyze the distinct assumptions, coordination variables, and scalability tradeoffs of the two approaches. Based on the insights obtained from this review, we formulate a hierarchical search-and-rescue (SAR) framework that combines role-level decision making, uncertainty-aware exploration, and graph-structured coordination. Finally, we describe the implementation of this SAR framework and discuss the qualitative behavior observed in the coded system. The resulting study shows that explicit organizational hierarchy and latent informational hierarchy are complementary tools for cooperative control rather than interchangeable alternatives.**

***Index Terms*—Multi-Agent Reinforcement Learning, Hierarchical Reinforcement Learning, Dec-POMDP, CTDE, Task Allocation, Consensus Learning, Search and Rescue**

I. Introduction

Cooperative multi-agent reinforcement learning (MARL) studies how multiple agents can learn coordinated behavior while acting in a shared environment. In many domains, especially robotics and distributed control, the main challenge is not only low-level actuation but also large-scale coordination under partial information. As the number of agents and interaction dependencies increases, flat MARL methods often struggle with delayed rewards, role ambiguity, and exploding joint state-action complexity.

Hierarchical MARL addresses this difficulty by introducing multiple levels of abstraction. A high-level controller may select roles, subtasks, or options, while a low-level controller handles primitive actions. Yet hierarchy itself is not unique. Different hierarchical methods place abstraction in different mathematical objects and thereby encode different views of coordination.

This report reviews two representative works. The first is ALMA, in which the authors model coordination through explicit subtask allocation and local subtask execution. The second is HC-MARL, in which the authors model coordination through hierarchical consensus variables learned from local observations. Although both methods are hierarchical, they are built around different principles. ALMA emphasizes *organizational hierarchy*, while HC-MARL emphasizes *informational hierarchy*.

The main purpose of this report is twofold. First, it provides a comparative theoretical review of these two papers. Second, it uses the insights gained from that review to motivate a theoretical framework for search-and-rescue (SAR), followed by an implementation of that framework. In this sense, the present work is not a re-derivation of ALMA or HC-MARL, but a review-driven synthesis that adapts the most relevant hierarchical ideas to a cooperative SAR setting. A general reward shaping rule using potential fields, explored in a lot of literature, is as described in the fig: 1.

II. Theoretical Background

A. Dec-POMDP Formulation

A cooperative MARL problem is naturally modeled as a decentralized partially observable Markov decision process (Dec-POMDP)

$$\mathcal{G} = (\mathcal{I}, \mathcal{S}, \{\mathcal{A}_i\}_{i=1}^N, P, r, \{\mathcal{O}_i\}_{i=1}^N, Z, \gamma), \tag{1}$$

where $\mathcal{I} = \{1, \ldots, N\}$ is the agent set, $\mathcal{S}$ is the global state space, $\mathcal{A}_i$ is the action space of agent i, P is the transition kernel, r is the shared reward, $\mathcal{O}_i$ is the local observation

space, Z is the observation function, and γ is the discount factor.

At time t, each agent receives

$$o_t^i = Z_i(s_t), \tag{2}$$

and the joint action is

$$\mathbf{a}_t = (a_t^1, \ldots, a_t^N). \tag{3}$$

A decentralized policy factorizes as

$$\boldsymbol{\pi}(\mathbf{a}_t \mid \mathbf{o}_t) = \prod_{i=1}^{N} \pi_i(a_t^i \mid o_t^i), \tag{4}$$

with objective

$$J(\boldsymbol{\pi}) = \mathbb{E}_{\boldsymbol{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, \mathbf{a}_t) \right]. \tag{5}$$

B. CTDE

Centralized training with decentralized execution (CTDE) allows learning with access to richer global information during training while preserving decentralized policies during execution. A centralized critic may use

$$Q^{\boldsymbol{\pi}}(s, \mathbf{a}), \tag{6}$$

but decentralized execution remains

$$a_t^i \sim \pi_i(\cdot \mid o_t^i). \tag{7}$$

This creates a gap between training-time information and execution-time information. HC-MARL directly addresses this gap through consensus augmentation, while ALMA addresses a different issue by structuring the task itself.

C. Hierarchical Reinforcement Learning

A standard option in hierarchical reinforcement learning is

$$\omega = (\mathcal{I}_\omega, \pi_\omega, \beta_\omega), \tag{8}$$

where $\mathcal{I}_\omega$ is the initiation set, π_ω is the intra-option policy, and β_ω is the termination condition [4]. In multi-agent settings, hierarchy may appear through role assignment, subtask decomposition, temporal abstraction, or latent shared context.

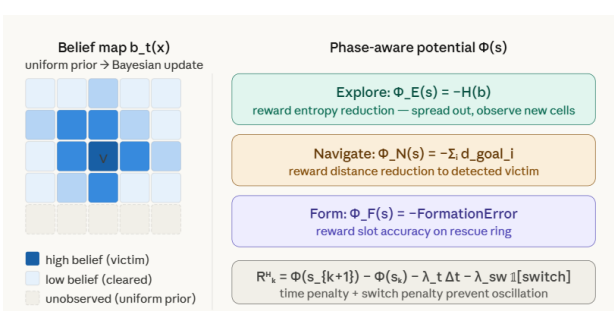

Fig. 1. Potential based reward the guarantees that the shaped reward doesn't change the optimal policy, which prevents reward hacking [3].

III. Review of ALMA

A. Core Formulation

In ALMA, the authors focus on *composite multi-agent tasks*, where the global objective can be decomposed into multiple subtasks [1]. Let the set of subtasks be

$$\mathcal{K} = \{1, \ldots, K\}. \tag{9}$$

The high-level decision is an allocation

$$b = \{b^a \mid a \in \mathcal{I}\}, \qquad b^a \in \mathcal{K}, \tag{10}$$

where b^a denotes the subtask assigned to agent a. Equivalently,

$$b_i = \{a \in \mathcal{I} : b^a = i\} \tag{11}$$

is the subset of agents assigned to subtask i.

The allocator is modeled through a value function over allocations,

$$Q_{\text{alloc}}(s, b), \tag{12}$$

with an ideal optimal allocation

$$b^*(s) = \arg\max_{b \in \mathcal{B}} Q_{\text{alloc}}(s, b). \tag{13}$$

Since the number of valid allocations grows as

$$|\mathcal{B}| = K^N, \tag{14}$$

the authors introduce a proposal distribution

$$q_\phi(b \mid s) \tag{15}$$

to sample promising allocations instead of exhaustively evaluating the full space.

B. Low-Level Execution

Once the high-level allocation is fixed, the authors construct local subtask states s_{b_i} containing only the entities relevant to subtask i [1]. For each assigned agent $a \in b_i$, the local utility is

$$Q_i^a(s_{b_i}, u^a; b), \tag{16}$$

and the corresponding low-level action is chosen by

$$u^a = \arg\max_{u'} Q_i^a(s_{b_i}, u'; b). \tag{17}$$

A subtask-level joint value is then written as

$$Q_i^{\text{tot}}(s_{b_i}, \mathbf{u}_{b_i}; b) = f_i\Big(\{Q_i^a(s_{b_i}, u^a; b)\}_{a \in b_i}\Big), \tag{18}$$

where f_i is a suitable mixing function.

C. Review Insight

From a review perspective, the key contribution of ALMA is that it makes hierarchy explicit at the *organizational* level. The paper treats coordination primarily as an allocation problem: before agents decide how to act, the system decides how responsibility should be distributed. This offers a strong inductive bias for tasks where the task structure itself is naturally decomposable.

At the same time, this strength also reveals the main assumption of the method. ALMA relies on meaningful subtask boundaries and on the idea that local execution can be conditioned on those boundaries without losing too much information. Thus, the paper is particularly well suited to structured environments in which the main difficulty is deciding *who should handle which part of the task*.

IV. Review of HC-MARL

A. Core Formulation

In HC-MARL, the authors begin from the CTDE setting and argue that decentralized agents often fail because local observations are too limited to support coordinated execution [2]. Their response is to learn a shared consensus variable from local observations.

Let $P_S(o_i)$ denote the student prediction over consensus categories and $P_T(o_j)$ denote the teacher prediction. The consensus objective is

$$L_S(\theta) = -\sum_{i=1}^{N}\sum_{j=1}^{N}\sum_{k} P_T(o_j)_k \log P_S(o_i)_k. \tag{19}$$

The consensus class assigned to agent i is

$$c_i = \arg\max_k P_S(o_i)_k. \tag{20}$$

To capture temporal structure, the authors extend this idea to sequences

$$x_i^m = \{o_{t_1}^i, o_{t_2}^i, \ldots, o_{t_m}^i\} \tag{21}$$

and define multi-level consensus variables

$$c_i^m = \arg\max_k P_S(x_i^m)_k. \tag{22}$$

These multi-scale representations are then aggregated through attention:

$$c_i^{\text{att}} = \text{MultiHead}(Q(c_i^m), K(c_i^m), V(c_i^m)). \tag{23}$$

The decentralized actor becomes

$$a_i \sim \pi_i(\cdot \mid o_i, c_i^{\text{att}}). \tag{24}$$

B. Review Insight

From a review perspective, the key contribution of HC-MARL is that it introduces hierarchy at the *informational* level rather than the organizational level. The paper does not ask which subtask each agent should receive. Instead, it assumes that the main coordination bottleneck lies in the inability of decentralized agents to reconstruct enough shared context from local views.

This makes HC-MARL theoretically attractive in environments where explicit subtask decomposition is weak or unavailable, but where agents still need a shared latent understanding to coordinate. However, the latent consensus variable is inherently less interpretable than an explicit allocation variable. Therefore, the paper trades interpretability for representational flexibility.

V. Comparative Review and Insights

The two reviewed papers place hierarchy in different mathematical objects:

- ALMA places hierarchy in the explicit allocation variable b.
- HC-MARL places hierarchy in the latent consensus variable c_i^{att}.

This leads to two different views of coordination:

1) **Allocative coordination:** the core problem is deciding which agent should handle which subtask.
2) **Informational coordination:** the core problem is enriching local observations with shared context.

From reviewing these works, several insights emerge:

- ALMA is more appropriate when the task has explicit compositional structure.
- HC-MARL is more appropriate when the task is difficult primarily because of partial observability and execution-time information gaps.
- ALMA is more interpretable because allocations are directly inspectable.
- HC-MARL is more flexible because it does not require pre-specified subtasks.
- Neither framework alone fully addresses settings where role allocation, uncertainty-aware exploration, and graph-structured multi-robot interaction all matter simultaneously.

This final point is especially important for search-and-rescue. It suggests that a realistic SAR system may require explicit task-level abstraction, partial-observation-aware learning, and topology-aware coordination at the same time.

VI. Search-and-Rescue Theoretical Framework

A. SAR as a Cooperative Hierarchical Problem

Motivated by the insights from the two reviewed papers, we formulate search-and-rescue (SAR) as a cooperative hierarchical control problem. The global state may include robot positions, obstacle map, victim map, hazards, and resource levels:

$$s_t = (\text{robots}, \text{map}, \text{victims}, \text{hazards}, \text{resources}). \tag{25}$$

Each robot receives only a local observation

$$o_t^i = Z_i(s_t). \tag{26}$$

Unlike a simple navigation task, SAR contains both allocative and informational difficulty. Robots may need to decide whether to *search*, *relay*, *assist*, or *support*, while also coordinating under incomplete observations. This motivates

TABLE I
COMPARATIVE REVIEW OF ALMA, HC-MARL, AND THE REQUIREMENTS OF SAR

Axis	ALMA	HC-MARL	SAR Need
Hierarchy locus	Explicit task allocation and execution	Hierarchical consensus representation	Task-level, informational, and topological hierarchy
Main coordination variable	Allocation vector b	Latent consensus c_i^{att}	Roles, beliefs, and graph context
Primary strength	Clear organizational structure	Better shared context under CTDE	Uncertainty-aware coordinated control
Interpretability	High	Moderate to low	Preferably high
Main assumption	Composite task decomposition exists	Shared latent structure can be learned	Dynamic, partially observed, many-agent environment
Main limitation	Weak when subtasks strongly overlap	Weak for explicit hard role assignment	Requires hybrid coordination mechanisms

a theoretical framework that combines role-level abstraction with uncertainty-aware coordination.

B. Belief-Based Exploration Objective

Let $b_t(x)$ denote a belief over victim presence at spatial location x. The belief entropy is

$$H(b_t) = -\sum_x b_t(x) \log b_t(x). \tag{27}$$

We define a mission potential

$$\Phi(s_t) = -H(b_t), \tag{28}$$

so that entropy reduction corresponds to improved situational awareness. At the high level, the reward between option decisions may be written as

$$R_k^H = \Phi(s_{t_{k+1}}) - \Phi(s_{t_k}) - \lambda_t(t_{k+1} - t_k) - \lambda_{sw}\mathbf{1}[z_k \neq z_{k-1}], \tag{29}$$

where z_k denotes the macro-level option at decision step k.

C. Hierarchical Policy

Inspired by task-level abstractions, we let the high-level controller choose a macro-role

$$z_k^i \sim \pi_H^i(z \mid h_{H,k}^i), \tag{30}$$

where z_k^i may correspond to search, relay, assist, or mitigation. The low-level controller then executes

$$a_t^i \sim \pi_L^i(a \mid h_{L,t}^i, z_k^i). \tag{31}$$

D. Graph-Structured Critic

Since SAR involves crowding, proximity, and local interaction, we model the multi-robot system through a graph

$$G_t = (V, E_t), \tag{32}$$

where an edge exists when two robots are within interaction radius r:

$$(i,j) \in E_t \iff \|p_i - p_j\| \leq r. \tag{33}$$

Let node features be

$$x_i = [o_t^i, z_k^i]. \tag{34}$$

A graph neural critic then computes

$$h_i^{(K)} = \mathrm{GNN}_\psi(x_i, \{x_j\}_{j \in \mathcal{N}(i)}), \tag{35}$$

followed by pooled context

$$h_{\mathrm{global}} = \mathrm{POOL}(\{h_i^{(K)}\}), \qquad V_\psi(s_t) = f_\psi(h_{\mathrm{global}}). \tag{36}$$

E. Topology-Conditioned Termination

In constrained environments, macro-behavior switching should depend on neighborhood structure as well as local state. A topology-aware termination function is written as

$$\beta_t^i = \sigma\Big(\mathrm{MLP}\Big(\mathrm{GNN}_\beta(o_t^i, \{o_t^j\}_{j \in \mathcal{N}(i)})\Big)\Big). \tag{37}$$

This SAR framework is therefore not a direct reproduction of either reviewed paper. Rather, it is a review-driven theoretical synthesis: explicit role abstraction is inspired by ALMA, while partial-observation-aware coordination is conceptually aligned with HC-MARL, and graph/topology mechanisms are added to match the demands of SAR.

VII. IMPLEMENTATION

Based on the above framework, we implemented a hierarchical MARL system for SAR. The implemented system includes:

- a high-level option or role-selection module,
- a low-level control policy conditioned on the selected role,
- a belief-aware objective that encourages entropy reduction,
- a graph-based centralized critic,
- a topology-conditioned termination rule for macro-behavior switching.

The implementation was designed to test whether the review-derived insights translate into coherent coordinated behavior in a simulated rescue setting. In particular, the implementation evaluates whether explicit role-level abstraction and graph-aware coordination improve search efficiency and reduce undesirable switching or congestion.

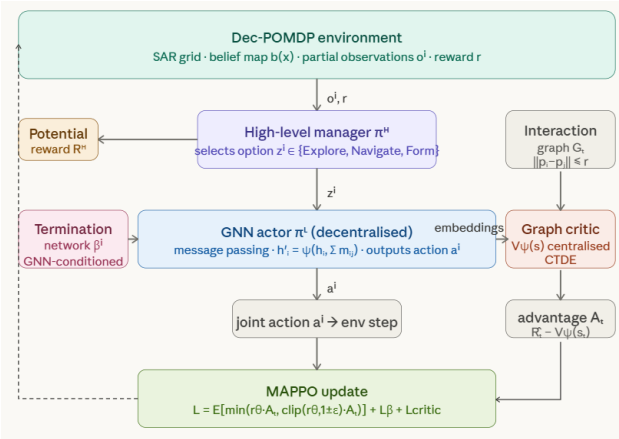

Fig. 2. Description of our SAR implemented architecture.

VIII. Experimental Setup and Qualitative Results

A. Experimental Setup

The implemented SAR framework as described in the fig: 2 was evaluated on representative cooperative scenarios such as:

1) large-area exploration,
2) clustered victim search,
3) narrow-passage coordination.

These scenarios were chosen because they stress different aspects of the framework: uncertainty reduction, role coordination, and topology-aware motion.

The implementation was compared against simpler baselines such as flat control or hierarchical variants with reduced structural components. The evaluation focused on qualitative and system-level indicators such as search efficiency, coordination smoothness, redundant motion, and the ability to avoid unstable switching in constrained regions.

B. Qualitative Results

The coded system showed that the theoretical framework is practically realizable. In exploration-heavy settings, the belief-based high-level objective encouraged broader coverage and more directed search behavior. In constrained topologies, the graph-aware components reduced unstable local behavior and improved coordination through bottleneck regions. The role-conditioned hierarchy also produced more structured team behavior than a purely flat controller.

Fig. 3. Representative result from the implemented SAR framework.

These observations are consistent with the central review insight of this report: explicit organizational abstraction and informational/topological coordination mechanisms address different aspects of cooperative complexity. In the SAR implementation, combining these ideas produced more coherent behavior than relying on any single abstraction alone.

IX. Conclusion

This report reviewed ALMA and HC-MARL as two distinct hierarchical MARL approaches and used the resulting insights to construct and implement a theoretical framework for search-and-rescue. The review showed that ALMA organizes coordination through explicit allocation, while HC-MARL organizes coordination through latent consensus. These choices correspond to different notions of hierarchy and different coordination bottlenecks.

Motivated by this comparison, the SAR framework developed in this report combines role-level abstraction, belief-aware exploration, graph-structured critics, and topology-conditioned switching. The resulting implementation demonstrates how ideas drawn from the reviewed literature can be synthesized into a new task-oriented hierarchical design. The broader conclusion is that hierarchical MARL should not be viewed as a single generic recipe. Instead, the most effective hierarchical structure depends on whether the main challenge is allocative, informational, topological, or some combination of all three.

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